

FINAL EXAMINATION

Coverage: Weeks 9–16 · Clustering & denoising, honest testing, portfolios, LLMs & RAG, reinforcement learning, causality & fairness, the frontier, and the project pipeline

Time allowed: 90 minutes · 40 questions × 2.5 points = 100 points

Instructions

- Choose exactly ONE answer (A–D) per question. No penalty for wrong answers.
- A simple calculator is allowed. All numbers are taken from the lecture hand-calculations.
- Name and student ID on the answer sheet. The answer key begins on a separate page — do not turn to it during the exam.

Part 1 · Structure & Honest Testing (Weeks 9–10)

1. K-Means on stock return features grouped names into low-beta and high-beta clusters (Week 9). K-Means assigns each point to:

- A) The cluster with the FARTHEST centroid
- B) The cluster with the NEAREST centroid, then recomputes centroids
- C) A random cluster each iteration
- D) The cluster with the most members

2. For two assets with correlation ρ , PCA of the 2×2 correlation matrix gives eigenvalues:

- A) ρ and $-\rho$
- B) $1 + \rho$ and $1 - \rho$
- C) 2ρ and 0
- D) 1 and ρ^2

3. The Marchenko–Pastur upper edge $\lambda_+ = (1 + \sqrt{(N/T)})^2$ is used to:

- A) Set the number of clusters
- B) Separate signal eigenvalues from pure-noise eigenvalues in a correlation matrix
- C) Compute transaction costs
- D) Annualize the Sharpe ratio

4. With $N = 100$ assets and $T = 400$ observations, $\lambda_+ = (1 + \sqrt{(100/400)})^2$ equals:

- A) 1.25
- B) 1.50
- C) 2.25
- D) 4.00

5. Purged cross-validation exists because financial labels:

- A) Are always independent
- B) Overlap in time, so nearby train/test samples leak information into each other
- C) Are too expensive to compute
- D) Never repeat

6. The embargo in purged CV additionally:

- A) Deletes the test set
- B) Skips a buffer AFTER the test window before training resumes
- C) Doubles the training data
- D) Randomly shuffles all rows

7. A strategy earns Sharpe 0.35 per month. Annualized ($\times\sqrt{12} \approx 3.464$), that is about:
- A) 0.42
 - B) 0.70
 - C) 1.21
 - D) 4.20
8. You test $N = 100$ junk strategies; Sharpe estimates have sd σ . The expected BEST Sharpe is approximately:
- A) 0
 - B) $\sigma \cdot \sqrt{(2 \ln 100)} \approx 3.03\sigma$
 - C) $\sigma/100$
 - D) $\sigma \cdot 100$
9. The Deflated Sharpe Ratio's essential job is to:
- A) Convert returns to log returns
 - B) Discount a backtest's Sharpe for the number of trials and non-normality
 - C) Maximize the in-sample Sharpe
 - D) Replace cross-validation
10. Which write-up sentence satisfies Week 10's standard?
- A) 'Our best variant achieved Sharpe 2.1.'
 - B) 'Sharpe 1.1 out-of-sample with purged CV; 12 variants were tried and are reported.'
 - C) 'The model felt robust.'
 - D) 'We stopped testing once results looked good.'

Part 2 • Portfolios & Language Models (Weeks 11–12)

11. Two assets each have variance 100 (in $\%^2 \cdot \text{units}$) and the equal-weight portfolio variance is $\sigma_p^2 = 125 + 100\rho$ – style toy from Week 11. At $\rho = 0$, diversification helps because:
- A) Variances add and double
 - B) Uncorrelated risks partially cancel, so portfolio variance is below the average asset's
 - C) Returns become risk-free
 - D) Beta becomes zero
12. Markowitz mean-variance weights are called 'unstable' (Week 11) because:
- A) The math has no solution
 - B) Tiny changes in estimated inputs can FLIP large allocations
 - C) They ignore expected returns
 - D) They require daily rebalancing by law
13. Inverse-Variance Portfolio (IVP) weights each asset:
- A) Equally
 - B) Proportional to $1/\text{variance}$ — calmer assets get more weight
 - C) By market cap
 - D) By recent performance
14. In the Week 11 hand-calc, HRP split the toy portfolio approximately:
- A) 50.0 / 50.0
 - B) 64.3 / 35.7
 - C) 90.0 / 10.0
 - D) 33.3 / 66.7
15. HRP's key structural idea is to:
- A) Invert the full covariance matrix directly
 - B) Use the correlation HIERARCHY (clusters) so weights never rely on a fragile matrix inverse

- C) Buy only the minimum-variance asset
- D) Ignore correlations entirely

16. In the Week 12 vector demo with $a = (2, 1)$ and $b = (4, 2)$, the cosine similarity is:

- A) 0
- B) 0.45
- C) 0.71
- D) 1.0

17. With softmax over scores $z = (2, 1, -1)$, the top choice received about 70.5% probability. RAISING the temperature to $T = 2$ made the distribution:

- A) Sharper (top $\approx 88\%$)
- B) Flatter (top $\approx 55\%$)
- C) Unchanged
- D) Deterministic

18. RAG (retrieval-augmented generation) reduces hallucination because the model:

- A) Is fine-tuned on more parameters
- B) Answers FROM retrieved documents it can cite, not from parametric memory alone
- C) Refuses all questions
- D) Uses a larger temperature

19. In the Week 12 mini-RAG, the query vector matched document d_2 with cosine ≈ 0.707 . The correct system behavior when NO document clears the threshold is to:

- A) Answer from memory anyway
- B) Return 'NOT FOUND' rather than fabricate — the honesty valve
- C) Lower the threshold until something matches
- D) Repeat the query verbatim

20. An LLM predicts:

- A) The true state of the world
- B) The next token given context — fluency is not a guarantee of truth
- C) Only numbers
- D) Database query results

Part 3 • Reinforcement Learning, Causality & Fairness (Weeks 13–14)

21. In the Week 13 choice between Grab $(10, 0, 0)$ and Invest $(-2, 0, 20)$, discounting with $\gamma = 0.9$ gives Invest ≈ 14.2 vs Grab 10. The indifference discount factor was $\gamma^* = \sqrt{0.6} \approx 0.775$, meaning:

- A) Below 0.775 you invest
- B) Above 0.775 the patient (Invest) plan wins
- C) γ never affects the choice
- D) Both plans always tie

22. With ϵ -greedy action selection at $\epsilon = 0.2$ and current best action A, the probability of playing A (two actions total) is:

- A) 0.8
- B) 0.9
- C) 0.5
- D) 0.2

23. The Q-learning update $Q \leftarrow Q + \alpha[r + \gamma \max_{Q'} Q' - Q]$ with $\alpha = 0.5$ moves the estimate:

- A) All the way to the new target
- B) Halfway from the old value toward the new target
- C) Away from the target

D) Only when the reward is positive

24. In the Week 13 execution mini (10,000 shares, impact $k \cdot t^2$), trading all at once cost 2,000 while TWAP in two halves cost 650. The saving comes from:

- A) Paying no fees
- B) Quadratic impact: splitting the order shrinks the t^2 penalty
- C) Prices always rising
- D) The broker's discount

25. RL suits trade execution (rather than direction prediction) because:

- A) Rewards are immediate and the action space is defined (how fast to trade)
- B) Markets are deterministic
- C) Labels are perfect
- D) Execution has no cost

26. In the Week 14 buzz demo, the naive buzz-return gap was +1.2pp but the WITHIN-regime gaps were 0 and 0. Conclusion:

- A) Buzz causes returns
- B) The regime confounded both — the entire signal was the hidden common cause
- C) More data is needed
- D) The gap was measurement error

27. With base 0.40, $f(I) = 0.30$, $f(D) = 0.70$, $f(D,I) = 0.65$, the Shapley value of Debt is:

- A) +0.30
- B) +0.325
- C) +0.35
- D) -0.075

28. The Shapley additivity property in that example verifies as:

- A) $0.40 + 0.325 - 0.075 = 0.65$ — contributions sum exactly to the prediction
- B) $0.40 \times 0.325 = 0.13$
- C) $0.70 - 0.30 = 0.40$
- D) No check is possible

29. SHAP's fundamental limitation, stressed repeatedly in Week 14:

- A) It only works for trees
- B) It explains the MODEL's arithmetic, not the world's causes
- C) It cannot handle more than two features
- D) It requires balanced classes

30. Both groups had 80 qualified applicants out of 100; approvals were 60 vs 40 (all qualified). The parity and opportunity gaps are:

- A) 10pp and 15pp
- B) 20pp and 25pp
- C) 0pp and 0pp
- D) 25pp and 20pp

Part 4 • The Frontier & The Final Project (Weeks 15–16)

31. A TFT's 'known future' inputs are things like:

- A) Tomorrow's realized returns
- B) Holidays and scheduled earnings dates — calendar facts certain in advance
- C) Other models' predictions
- D) Random noise features

32. For past = (10.0, 10.5, 9.8, 12.0, 11.2, 13.5), recency attention weights (1..6)/21 give a forecast of $245.4/21 \approx 11.69$ versus a flat average of 11.17. The point of the exercise:

- A) Attention changes the data
- B) The WEIGHTS are the model — a real TFT learns them from data
- C) Flat averages are always best
- D) The last day should be ignored

33. In the 4-node GNN hand-calc (links A-B, A-C, B-C, C-D; shock 5 at A), node D first became nonzero (0.42) in round 2 because:

- A) Its degree is 1
- B) D is two hops from A and each round passes news one edge
- C) The 0.5 blend is too small
- D) D was shocked directly

34. An AI agent differs from a chatbot chiefly because it:

- A) Has more parameters
- B) Uses tools in a think-act-observe loop, so it can DO things (and do them wrongly)
- C) Never hallucinates
- D) Requires no LLM

35. Agent EV mini: right = +10, wrong = -20, tool call = 1. Memory has $P = 0.6$; RAG has $P = 0.95$. Their EVs are:

- A) -2.0 and +7.5 — one tool call flips the sign
- B) +6.0 and +9.5
- C) 0 and 0
- D) -8.0 and -1.0

36. When the penalty for a wrong answer drops to $L = 2$, the EV ranking changed so that:

- A) Memory became positive (+5.2) and plain RAG won (+8.4)
- B) Nothing changed
- C) Search+RAG won by more
- D) All EVs went negative

37. The five stages of the final-project pipeline are:

- A) Data, model, honest test, explanation & fairness, presentation
- B) Model, model, model, test, deploy
- C) Collect, clean, train, retrain, forget
- D) Idea, code, backtest, leverage, launch

38. Fraud audit: 1,000 rows, 20 frauds; TP 8, FP 2, FN 12, TN 978. Accuracy is 98.6% but F1 is:

- A) 0.80
- B) 0.533
- C) 0.40
- D) 0.986

39. A trading write-up claims 12% a year (edge 30bp, cost 12bp, 40 trades, vol 10%). The honest headline numbers are:

- A) Net 7.2% and net Sharpe 0.72
- B) Gross 12% and Sharpe 1.2
- C) Net 3.6% and Sharpe 0.36
- D) Breakeven 12bp

40. On the project rubric (weights .20/.15/.30/.20/.15), the 'fancy model, thin testing' draft scored 5.20 while the 'simple model, honest work' draft scored 8.05. The structural reason:

- A) The grader disliked fancy models

- B) Test + explain carry 50% of the weight — honesty is half the grade
- C) The simple model had higher accuracy
- D) Presentation was weighted most